c. Gynostegium:Connation product of stamens and stigma is called **gynostegium**. Example: *Calotropis* and Orchidaceae.

d. Pollinium: Pollen grains are fused together as a single mass



Figure 4.25: (b)
Gynostegium



Figure 4.25: (c)
Pollinium

4.4.2 Arrangement of stamens relate to length of stamens:

1. **Didynamous (di-two, dynamis-strength)**: Four stamens in which two with long filaments and two with short filaments. Example: Lamiaceae, *Ocimum*. If all four stamens are in two equal pairs then the condition is called **didynamous.**



Figure 4.26: (a)
Didynamous

2. **Tetradynamous(tetra-four)**: Six stamens of which four with long filaments and two with short filaments. Example: Brassicaceae, (*Brassica*).

3. **Heterostemonous**: stamens are of different lengths in the same flower. Example: *Cassia, Ipomoea.*



Figure 4.26: (b)
Tetradynamous



Figure 4.26: (c)
Heterostemonous

4.4.3 Stamen insertion

1.Inserted: Shorter than the corolla tube and included within. Example: *Datura*.

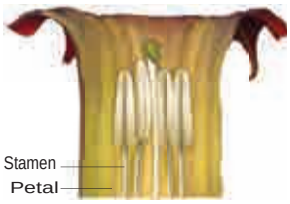


Figure 4.27: (a)
Inserted



Figure 4.27: (b)
Exserted

2.Exserted:Longer than the corolla tube and project out.Example: *Mimosa, Acacia arabica*

The number of whorls of stamens present in a flower is called **stamen** cycly. Two major types are 1.**uniseriate**,a single whorl of stamens and 2.**biseriate,**two whorls of stamens.

4.4.4 Anther types

1. Monothecal: One lobe with two microsporangia. They are kidney shaped in a cross section. Example: Malvaceae



Figure 4.28: (a)
Monothecal



Figure 4.28: (b)
Dithecal